1. Project Overview

Technology: Flutter (Dart) — multi-module School ERP application. App ek single Dashboard Shell se start hota hai, jisme ek permanent Sidebar (17 menu items) hoti hai. Sidebar se select karne par Expanded content area mein respective module load hota hai — kai modules (Fees, Examinations, Transport, Hostel, HR & Payroll, Inventory, Settings) apne aap mein complete self-contained mini-apps hain, apni khud ki internal navigation ke sath.

High-Level Flow

→ App Launch (main.dart) → EduSmartApp (MaterialApp)

→ Dashboard Screen (Home Shell) → Sidebar (left) + Top Header + Content Area

→ Sidebar Selection → Content Area mein respective Module load hota hai

→ Har Module ke andar apni Screens / Tabs / Dialogs / Data Tables hote hain

Sidebar — 17 Main Menu Items

- 1. Dashboard
- 2. Students
- 3. Teachers
- 4. Parents
- 5. Attendance
- 6. Fees
- 7. Examinations
- 8. Homework (planned - placeholder)
- 9. Library (planned - placeholder)
- 10. Transport
- 11. Hostel
- 12. HR & Payroll
- 13. Inventory
- 14. Reports (planned - placeholder)
- 15. Notifications (planned - placeholder)
- 16. Settings
- 17. System Setup (planned - placeholder)

2. Main App Shell — Dashboard (Home Screen)

App launch se sabse pehle yahi shell khulti hai. Isme ek fixed Sidebar (left) + Top Header + scrollable Dashboard body hota hai. Sidebar se hi baaki 16 modules select hote hain.

▸ Sidebar (left navigation) — widgets/sidebar.dart

- Elements:
  - Logo block (EduSmart icon + 'EduSmart / School ERP' text)
  - 'MAIN MENU' section label
  - 17 Menu Tiles (list above) — har tile: icon + label, selected-state highlight, collapsed-mode icon-only view
  - Notifications menu-item par badge counter (8)
  - Light/Dark Mode toggle switch (bottom)

- Collapse/Expand button (floating circular button on sidebar edge)

▶ Top Header — widgets/top_header.dart

- Elements:
  - Welcome text: 'Welcome back, Admin! 👋' + subtitle
  - Search Bar (hint: 'Search (Students, Teachers, Fees etc.)') + circular search-icon button
  - Academic Year Dropdown (shows '2024-2025' + dropdown arrow)
  - Notification Icon Badge (count 6)
  - Mail/Message Icon Badge (count 5)
  - Profile Section (avatar circle, 'Admin User' name, 'Super Admin' role, dropdown arrow)

▶ Dashboard Body — screens/dashboard_screen.dart

- Stat Cards Row (5 cards) — widgets/stat_cards_row.dart:
  - Total Students — 2,568 (+12.5% from last month)
  - Total Teachers — 245 (+8.3% from last month)
  - Total Users — 1,987 (+10.2% from last month)
  - Today's Attendance — 85.6% (-2.1% from yesterday)
  - Monthly Collection — ₹18,65,250 (+15.8% from last month)
- Row 2 — 4 cards:
  - Fee Collection Card — title 'Fee Collection Overview'; sub-labels Collected / Pending / Overdue
  - Admission Chart Card — title 'Students Admission' (chart)
  - Quick Access Hub Card — title 'Quick Access Hub'; 6 action buttons: Add Student, Add Teacher, Take Attendance, Collect Fees, Generate Report, Create Notice
  - Upcoming Events Card — title 'Upcoming Events' (event list)
- Row 3 — 3 cards:
  - Attendance Heatmap Card — legend: 90% and above / 75%-89% / 50%-74% / Below 50%
  - Recent Activities Card — title 'Recent Activities' (activity feed)
  - Fee Due Alerts Card — title 'Fee Due Alerts' (alert list)
- Bottom Shortcut Bar — widgets/bottom_bar.dart:
  - 'Shortcuts' chip
  - Quick icon shortcuts: Calendar, Students (Groups), Library, Fees (₹), Reports (bar chart)
  - Floating utility buttons (right side): Settings, Fullscreen, AI/Sparkle (highlighted)
  - Footer copyright text: '© 2024 EduSmart School ERP. All rights reserved.'

3.1 Module 2 — Students

▶ Students Dashboard Screen

- Stat / KPI cards:
  - Total Students
  - Boys
  - Girls
  - Present Today
  - New Admissions
- Quick Action tiles:
  - Add Student — 'Add new student record'

  - Import Students — 'Import from Excel/CSV'
  - Export Students — 'Export student data'
  - Send Notice — 'Send notification to parents'
  - Send SMS — 'Send SMS to parents'
  - Generate ID Cards — 'Generate student ID cards'
- Charts / Info cards:
  - Student Growth (chart)
  - Class-wise Students (chart)
  - Boys vs Girls (donut chart)
  - Attendance Overview
  - Transport Users
  - Hostel Residents
  - Document Verification
  - ID Cards Generated
- Other dashboard widgets:
  - At-Risk Students Card — risk levels: High Risk / Medium / Low
  - Categories Card
  - Today Highlights Card
  - Top Performers Card
  - Upcoming Needs Card
- Students List Section (data table):
  - Toolbar buttons: Add Student, Import Excel, Export PDF, Reset
  - Search field (hint: 'Search by name, admission no., roll no...')
  - Table columns: Photo, Admission No., Name, Class, Section, Mobile No., Parent/Guardian, Status, Action
  - Row action menu ('...')
  - Delete-confirmation dialog: 'Delete student?' with Cancel / Delete buttons

▶ Add / Edit Student Screen (multi-step form)

- Elements:
  - Multi-step wizard navigation — Back / Next buttons
  - Toggle: 'Uses School Transport'
  - Toggle: 'Lives in Hostel'
  - Document attachment upload control

▶ Student Profile Screen (tabbed)

- Tabs (10):
  - Overview
  - Academic
  - Attendance
  - Fees
  - Documents
  - Parents
  - Transport

  - Hostel
  - Examinations
  - Communication
- Actions:
  - Record Payment (fees tab action)

3.2 Module 3 — Teachers

▶ Teachers Dashboard Screen

- Stat cards:
  - Total Teachers
  - Departments
  - Present Today
  - On Leave
  - New This Month
- Charts / cards:
  - Department Wise Teachers (chart)
  - Upcoming Birthdays (Birthday Card)
- 'Teaching Tools & Important Links' section (icon tiles):
  - My Timetable / View Schedule
  - Class Allocation / View Classes
  - Subject Allocation / View Subjects
  - Attendance Report / Track Progress
  - Leave Management / Apply-Approve
  - Exam Schedule / Upcoming Exams
  - Performance Report / View Reports
  - Syllabus Coverage
- 'Additional Important Information' section:
  - Info tiles (summary counts)
- Teacher List Panel (data table):
  - Filter chips: All / Active / On Leave
  - Search field (hint: 'Search by name, email or subject...')
  - Filter controls: Department, Apply, Clear
  - Table columns: EMP ID, TEACHER, DEPARTMENT, SUBJECT, CONTACT, STATUS, ACTION
  - Row actions: View Profile, Edit, Remove (with 'Remove teacher?' confirm dialog)
  - Pagination controls
- Dialogs:
  - Teacher Form Dialog (Add/Edit teacher)
  - Quick Action dialogs: My Timetable (day chips + periods), Leave Management (Approve/Reject requests, hint 'Reason for leave...'), Simple List body, Progress List body
- Top Header (module-specific):
  - Search field (hint: 'Search here...')
  - Academic session switcher: 2023–24 (archived) / 2024–25 (current) / 2025–26
  - Notifications panel

  - Messages panel
  - Account/Profile dialogs — toggles: Auto-Sync Timetable, Email Notifications, SMS Alerts; Log-out confirmation

3.3 Module 4 — Parents

Yeh module apni khud ki complete App Shell (sidebar + header) ke sath embed hai — 14 sections:

▶ Parents App Shell — 14 sidebar sections

- Sections:
  - Dashboard
  - My Children
  - Attendance
  - Academics
  - Fees & Payments
  - Examinations
  - Homework
  - Timetable
  - Transport
  - Library
  - Notices
  - Events
  - Teacher Communication
  - Settings
- Other screens:
  - Child Selector (switch between children)
  - Add Child dialog

3.4 Module 5 — Attendance

▶ Attendance Dashboard Screen

- KPI cards (6):
  - Present Today
  - Absent Today
  - Late Students
  - Attendance Rate
  - Monthly Average
  - Teachers Present
- Analytics Card:
  - Range Tabs (period filter)
  - Area/Trend Chart
- Distribution Card:
  - Attendance distribution donut chart
- Attendance Table (main data grid):
  - Toolbar: search (hint 'Search by name, roll no...'), filter chips, Clear-all-filters chip
  - Bulk Action Bar (multi-select actions)

  - Column headers + per-row Status Radio Group
  - Row actions: Edit Status, Notify Parent, View History
  - 'All Status' filter, 'Only check-in at/after 9:00 AM', 'Only students with remarks' filters
  - Empty-state view
  - Pagination controls (page buttons + nav buttons)
- Filter Bar:
  - Dropdown filter controls
  - Refresh button
- Dialogs:
  - Mark Attendance Dialog — student search (hint 'Search student by name or roll no...'), remarks field
  - Import Attendance Dialog — dotted file drop-zone (Excel/CSV import)
  - Export Report Dialog

3.5 Module 6 — Fees (self-contained mini fee-ERP)

Apni khud ki tab-bar navigation ke sath — 8 sections:

▶ 1) Dashboard

- KPI cards:
  - Total Collected (Session)
  - Outstanding Dues
  - Students Overdue
  - Net Revenue
  - Total Expenses
  - 'Live' status chip
- Charts / cards:
  - Collection Trend — Last 14 Days
  - Payment Mode Split
  - Recent Activity list
  - Session Collection Progress
- Actions:
  - Collect Fee (button)

▶ 2) Students & Fees

- Students List Screen:
  - Search (hint: 'Search by name, admission no., roll no. or phone…')
  - Filters: Class (All classes / Class-wise), Fee Status (All / Fully Paid / Has Dues / Overdue)
- Student Profile Screen:
  - Fee Head Breakdown card — Total Fee / Paid / Balance Due / Scholarship / Transport / Hostel / Optional
  - Parent / Guardian info card
  - Installment Planner (timeline)
  - Payment History card
  - Collect Fee action

▸ 3) Fee Collection

- Elements:
  - Select Student (search)
  - Select Installment
  - Fee Heads panel
  - Payment Summary
  - 'Collect & Generate Receipt' button
  - 'Full Profile' link

▸ 4) Defaulters

- Elements:
  - Defaulter bucket cards (aging buckets)
  - Bulk Reminder button

▸ 5) Reports

- Report types:
  - Collection vs Outstanding by Class (Class-wise Report)
  - Head-wise Collection Report
  - Payment Mode Report
  - Day Book / Cash Book Report
- Actions:
  - Export CSV
  - Print Summary
  - Copy to clipboard

▸ 6) Finance

- KPI cards:
  - Fee Income
  - Other Income
  - Total Expenses
  - Net Balance
- Charts / ledgers:
  - Category-wise Expenses (chart)
  - Income vs Expense — Last 6 Months (chart)
  - Income Ledger list
  - Expense Ledger list
- Dialogs:
  - Add Income dialog
  - Add Expense dialog

▸ 7) Day Closing

- Counter Controls card:
  - Opening Cash
  - Cash Collected Today
  - Online Collected Today

  - Expected Closing Cash
- Actions:
  - Close Counter
  - Reopen Counter
  - Update Opening Cash
  - Print Day Summary
- Other:
  - Today's transaction list

▶ 8) Audit Log

- Elements:
  - Action log entries — action, detail, user, IP address

▶ Shared / cross-module

- Receipt Screen:
  - School Info header
  - 'STUDENT DETAILS' section
  - 'PAYMENT SUMMARY' section
  - Actions: Print, PDF, Email, WhatsApp, Cancel Receipt (with confirm dialog)
- Top bar:
  - Notifications button
  - Global Search / Command Palette (Ctrl+K style)
  - Role Switcher
- Command Palette:
  - Universal search across students, receipts, and pages

3.6 Module 7 — Examinations (self-contained mini exam-ERP)

Sidebar mein 27 sections hain — sabse bada module:

▶ 1) Dashboard

- KPI cards:
  - Total Students
  - Eligible Students
  - Upcoming Exams
  - Books Collected
  - Evaluation Pending
  - Evaluation Completed
  - Results Published
  - Malpractice Cases

▶ 2) Exam Management

- Elements:
  - Exam list table — columns: Exam Name, Class, Type, Subjects, Status, Actions
  - Exam Form Dialog — Save Exam / New Examination / Delete exam (confirm)

▶ 3) Exam Calendar / Timetable

- Elements:
  - Full calendar/timetable view of scheduled exams

▶ 4) Student Eligibility

- Table columns:
  - Roll No.
  - Student Name
  - Class
  - Status (Eligible/Not Eligible/Conditional)
  - Reason
  - Action

▶ 5) Question Bank

- Table columns:
  - Question
  - Subject
  - Chapter
  - Type
  - Difficulty
  - Marks
  - Actions
- Actions:
  - Add Question dialog

▶ 6) Question Paper Generator

- Elements:
  - Set-wise paper builder (Set A/B/C…)
  - Question list with marks & difficulty
  - 'Generate Paper' button

▶ 7) Question Paper Security & Approval

- Table columns:
  - Paper
  - Subject
  - Questions
  - Marks
  - Security
  - Status
  - Action

▶ 8) Seating & Invigilator Management

- Elements:
  - Seat plan generator
  - Invigilator duty roster
  - Legend: Available/Occupied seats

▸ 9) Exam Hall Management

- Elements:
  - Live attendance grid
  - Answer-book distribution/collection tracking
  - Present/Absent, Books Distributed/Collected counters

▸ 10) Barcode / Security Center

- Elements:
  - Anonymous ID & Mapping table (Subject, Status)
  - 'Generate Codes' / 'Print / Download All (A4)' actions

▸ 11) Answer Book Custody

- Table columns:
  - Anon ID
  - Subject
  - Status
  - Advance Status
  - History
  - 'Reveal identity' action

▸ 12) Anonymous Evaluation

- Elements:
  - Blind Evaluation card (no student identity shown)
  - Marks entry per question (with max marks)
  - Examiner comment field
  - Save Draft / Submit & Lock Marks / Next Book

▸ 13) Moderation & Marks Lock

- Table columns:
  - Book ID
  - Subject
  - Marks Awarded
  - Status
  - Approve / Reject actions

▸ 14) Result Processing

- Table columns:
  - Student
  - Result
  - Percentage
  - Grade
  - Rank
  - Stage
  - Passed/Failed counts
- Actions:

- Compute All Results

▶ 15) Grading Engine (configuration)

- Elements:
  - Grade scale table (Add Grade)
  - 'Save Grade Scale' action

▶ 16) Report Card Designer

- Elements:
  - Template list ('Add New Template')
  - Template editor
  - 'Save Template' action

▶ 17) Revaluation / Rechecking

- Table columns:
  - Student
  - Subject
  - Original marks
  - Revised marks
  - Reason
  - Status
- Actions:
  - New Revaluation Request
  - Approve / Reject

▶ 18) Admit Card Management

- Elements:
  - Generate / preview / print / export admit cards

▶ 19) Certificate & Transcript Management

- Table columns:
  - Student
  - Class
  - Reference / Verification ID (auto CERT-xxxx)

▶ 20) Exam Reschedule / Cancellation

- Elements:
  - Current date vs New date
  - 'Reschedule Exam' / 'Confirm Reschedule' / 'Cancel exam' dialogs

▶ 21) Substitute Invigilator Management

- Table columns:
  - Exam
  - Date
  - Time
  - Room

- Subject
- Original Invigilator
- Substitute
- Status

▸ 22) Digital Exam Instructions

- Elements:
  - Instruction set list
  - 'Add Instruction' / 'Save Instructions' actions

▸ 23) Malpractice / Unfair Means Management

- Table columns:
  - Student
  - Type
  - Room/Seat
  - Description
  - Status
  - Action
- Actions:
  - Report Malpractice Incident

▸ 24) Special Examination Accommodation

- Elements:
  - Accommodation records list

▸ 25) Analytics & Reports

- Elements:
  - Performance / evaluation-progress / operational analytics charts

▸ 26) Notifications / Alerts Center

- Elements:
  - Notification list (body + audience)
  - 'Create Notification' / 'Send Now' dialog

▸ 27) Student & Parent Portal

- Elements:
  - Read-only portal preview (per student, by Roll No.)

▸ 28) Archive (previous examinations)

- Elements:
  - Searchable, immutable archive list

▸ 29) Backup & Data Security

- Elements:
  - 'Backup Now' action
  - Backup history

▸ 30) Roles & Permission Management

- Table columns:
  - Role
  - Permissions matrix

▸ 31) Audit Logs

- Table columns:
  - Timestamp
  - User
  - Type
  - Details — immutable log of every sensitive action

3.7 Module 10 — Transport

Single dashboard-page app; sidebar sections: Routes, Vehicles, Drivers, Students, Attendance, Pickup & Drop, Transport Fees, Notifications, Messages, Announcements, Reports, Analytics, Export Data.

▸ Transport Dashboard

- Elements:
  - Welcome banner
  - Quick Actions panel
  - Route Summary table — columns: Route Name, Vehicle No., Driver, Students, Status, Performance
  - Upcoming Trips table — columns: Trip Type, Time, Date, Mode
  - Transport Fee Collection card — Amount, Transactions (Txns), Pending Amount, This Month
  - Attendance donut — Picked Up / Yet to Pickup / Absent counters

3.8 Module 11 — Hostel

Sabse detailed sidebar tree (13 top-level groups, har ek ke andar sub-items):

▸ 1) Dashboard

- Stat cards:
  - Total Students
  - Total Rooms
  - Blocks
  - Present Today
  - Pending Payments
- Cards:
  - Room Occupancy Overview
  - Recent Payments
  - Hostel ID Card Generator ('Generate for This Student')
  - Quick Actions (incl. Logout)
  - Notices feed: 'Room 205 Maintenance', 'All Rooms Cleaned', '3 Student Leaves Pending', '2 Mess Complaints Pending'

▸ 2) Student Management

- Sub-sections:

- All Students
- Add Student
- Student Profiles
- Student Documents

▶ 3) Hostel Management

- Sub-sections:
  - Blocks
  - Floors
  - Rooms
  - Beds
  - Room Allocation
  - Vacant Beds

▶ 4) Attendance

- Sub-sections:
  - Daily Attendance
  - Attendance History
  - Attendance Reports

▶ 5) Leave Management

- Sub-sections:
  - Leave Requests
  - Pending Requests
  - Approved Leaves
  - Rejected Leaves

▶ 6) Fees & Payments

- Sub-sections:
  - Fee Dashboard
  - Fee Structure
  - Pending Fees
  - Payment History
  - Receipts

▶ 7) Mess & Food

- Sub-sections:
  - Today's Menu
  - Weekly Menu
  - Meal Attendance
  - Mess Expenses
  - Food Inventory

▶ 8) Complaints

- Sub-sections:
  - All Complaints

- Pending Complaints
- Resolved Complaints

▶ 9) Maintenance

- Sub-sections:
  - Maintenance Requests
  - Pending Maintenance
  - Completed Maintenance
  - Maintenance Staff

▶ 10) Visitors

- Sub-sections:
  - Visitor Log
  - Today's Visitors
  - Visitor History

▶ 11) Notices

- Sub-sections:
  - All Notices
  - Create Notice
  - Notice Categories

▶ 12) Staff Management

- Sub-sections:
  - Staff List
  - Add Staff
  - Staff Roles
  - Staff Attendance

▶ 13) Reports

- Sub-sections:
  - Student Reports
  - Hostel Reports
  - Attendance Reports
  - Fee Reports
  - Leave Reports
  - Mess Reports
  - Maintenance Reports

▶ 14) Settings

- Sub-sections:
  - Hostel Settings
  - Academic Year
  - User Management
  - Roles & Permissions
  - Notifications

- General Settings

3.9 Module 12 — HR & Payroll

Sidebar 5 groups mein organised hai:

▶ Group A — Employee Management

- Sub-sections:
  - Dashboard
  - Employees
  - Departments
  - Designations
  - Attendance
  - Leave Management
  - Shift Management
  - Appraisals

▶ Group B — Payroll Management

- Sub-sections:
  - Salary Structure
  - Payroll Processing
  - Allowances & Deductions
  - Loans & Advances
  - Payslips
  - Tax Management

▶ Group C — Recruitment

- Sub-sections:
  - Jobs
  - Candidates
  - Interviews
  - Offers

▶ Group D — Reports & Analytics

- Sub-sections:
  - HR Reports
  - Payroll Reports
  - AI Analytics

▶ Group E — Others

- Sub-sections:
  - Notifications
  - Settings
  - User Profile

▶ Dashboard screen (detail)

- KPI cards:

  - Total Employees
  - Departments
  - Attendance Today
  - On Leave Today
  - Average Attendance
  - Total Payroll (May)
  - Pending Payments
  - Absent Today
  - On Leave
- Charts / cards:
  - Employee Overview
  - Attendance Overview (This Month)
  - Payroll Summary
  - Payroll Trends (Last 6 Months)
  - Top Departments by Headcount
  - Recent Joinings
  - Upcoming Birthdays
  - Quick Actions

3.10 Module 13 — Inventory

▶ Inventory Dashboard (single page)

- Header Section:
  - Profile menu
  - Settings menu
- Toolbar Section:
  - Search (hint: 'Search items, categories, brands...')
  - 'Add New Item' button
  - 'Export Report' button
  - 'More Actions' menu
  - Import Items
  - Print Table
  - Refresh / Clear Filters
- KPI Section:
  - Total Items
  - Total Value
  - Low Stock Items
  - Out of Stock
  - Reorder Soon
  - Active Categories
  - Total Categories
  - Not Available
- Inventory Overview Section:
  - Available Items

  - Issued Items
  - Reserved Items
  - Damaged Items
  - Expired Items
  - Suppliers
- Other sections:
  - Category Wise Items
  - Low Stock Items list
  - Inventory Alerts
  - Quick Actions
  - Inventory Items table (with filters)
  - Stock Activity — Recent Stock In / Recent Stock Out
  - Summary Section — Total Items, Total Quantity, Total Value, Active Categories, Active Suppliers
- Dialogs:
  - Add / Edit Item dialog (with Auto reorder-status option)
  - Add Supplier dialog
  - Confirm Delete Item dialog
  - Item Detail dialog
  - Stock Move (In/Out) dialog

3.11 Module 16 — Settings

▶ ERP System Settings screen

- Category Cards (6):
  - System Settings — Global configuration
  - Security — Access & protection
  - Users & Roles — Manage access
  - Backup — Data safety
  - Notifications — Alerts & messages
  - Appearance — Look & feel
- General Settings Form fields:
  - Organization / Institute Name
  - Academic Year / Session
  - Language
  - Time Zone
  - Date Format
  - Currency
  - Items Per Page
  - Theme
- Other elements:
  - Settings Header
  - Toggle switches (Setting Toggle)
  - Save Settings Bar
  - Toast notifications

5. Module Summary Table

#	Module	Status	Screens / Sections Count
1	Dashboard (Home Shell)	Active	1 main screen (Sidebar+Header+Body)
2	Students	Active	3 screen groups
3	Teachers	Active	1 screen groups
4	Parents	Active	1 screen groups
5	Attendance	Active	1 screen groups
6	Fees (self-contained mini fee-ERP)	Active	9 screen groups
7	Examinations (self-contained mini exam-ERP)	Active	31 screen groups
8	Transport	Active	1 screen groups
9	Hostel	Active	14 screen groups
10	HR & Payroll	Active	6 screen groups
11	Inventory	Active	1 screen groups
12	Settings	Active	1 screen groups